TECHNICAL DESIGN

BeReal Passport & Community Boards — Technical Design

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> *Independent concept created for a UI/UX Design & Product Management Externship. Not affiliated with or endorsed by BeReal.*

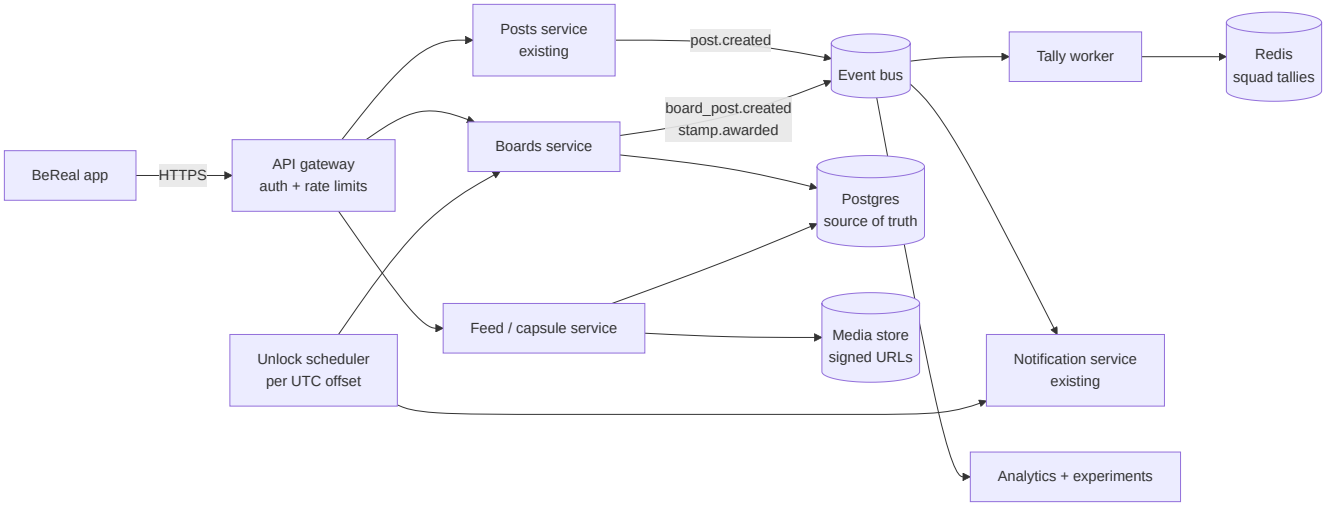
Summary

This page proposes how BeReal Passport & Community Boards could be built **on top of BeReal's existing systems**, adding as little as possible. The core idea: **a board post is a tag on an existing BeReal post, not a new kind of content.** The photo, camera, upload and moderation all stay as they are. The feature adds four things:

1. **Data model:** squads, boards, board memberships, board posts and stamps
2. **Scheduling:** boards unlock at 12:00 AM *local* time for every user
3. **Squad Stamp Tally:** a squad leaderboard that's fast to read and always correct
4. **Privacy:** squad-only visibility, enforced on the server

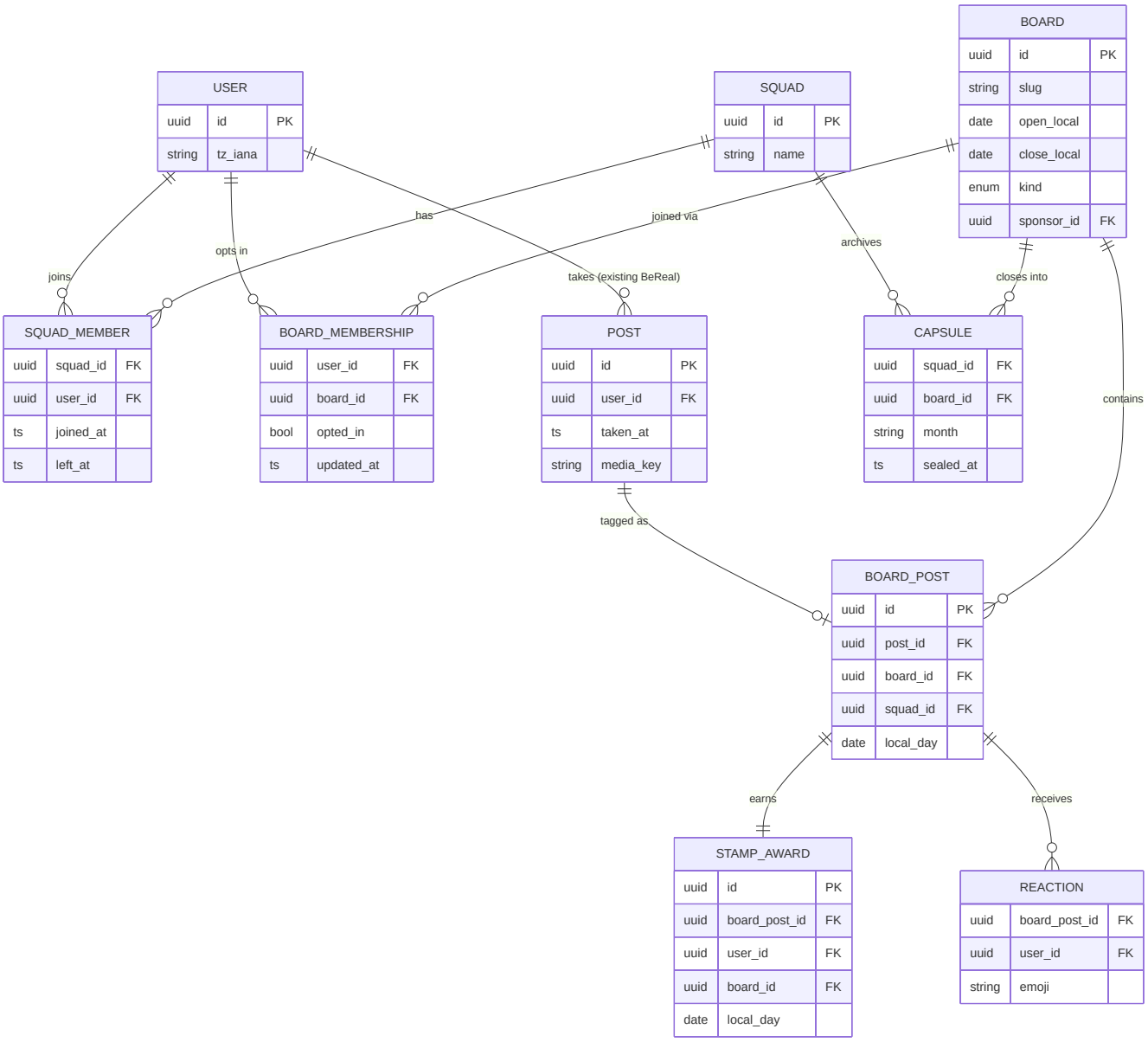
BeReal's real architecture isn't public. Services marked "existing" are assumed, and the numbers here are planning assumptions.

Architecture



Component	New or existing	Responsibility
Boards service	New	Boards, opt-in and opt-out, tagging posts to boards, awarding stamps
Unlock scheduler	New	Opens boards and sends the 12:00 AM alert per timezone
Tally worker	New	Keeps each squad's Stamp Tally up to date
Feed / capsule service	New	Board feeds, the Passport, monthly capsules; enforces squad-only access
Posts, notifications, media, moderation	Existing	Unchanged; boards reuse them

1. Data model



Key decisions:

Decision	Why
`BOARD_POST` points to an existing `POST` rather than copying it	The photo stays identical to a regular BeReal (the PRD's authenticity guardrail), and deleting the post removes it from the board automatically
Boards store local *dates* (`open_local` , `close_local`), not timestamps	"October 1" means midnight in *each user's* timezone, not one global moment (see section 2)
`BOARD_MEMBERSHIP` is per user, per board	Supports opting in or out board by board, as in Settings → Opt-in / Opt-out - Community Board

Decision	Why
`BOARD_POST` records the `squad_id`	Every read can be checked against squad membership cheaply (see section 4)
Unique constraint on `STAMP_AWARD (user_id, local_day)`	**One stamp per user per day**, matching BeReal's one-photo-a-day rhythm and preventing stamp farming
Sponsored boards are boards with a `sponsor_id`	No separate system; stamp art and taglines are configuration, not code

Key API endpoints:

Endpoint	Purpose
`GET /boards/today`	Live boards for the user's local date, with lock state and squad post counts
`PUT /boards/{id}/membership`	Opt in or out of a board
`POST /posts/{id}/board`	Tag today's post to a board and award the stamp. Requires an **idempotency key**, so a retry after "That didn't go through" never double-stamps
`GET /passport` · `GET /squads/{id}/tally`	Stamp gallery and Squad Stamp Tally
`GET /boards/{id}/feed` · `GET /capsules/{id}`	Squad-only board feed and monthly capsule

2. Scheduling the 12:00 AM unlock

The problem: boards open at **12:00 AM in each user's own timezone**. One global unlock would open boards at 3 PM for some users and 4 AM for others.

The approach: unlock by UTC offset, not by user.

1. **Each user has a timezone** (`tz_iana` , e.g. `America/New_York`), taken from the device.
2. **Users are grouped by their current UTC offset.** There are fewer than 40 offsets worldwide, including half-hour and 45-minute ones, so the scheduler has fewer than 40 jobs a night, not millions.
3. **The scheduler fires at each offset's midnight** and does two things: - marks the day's boards as open for that offset group - queues the 12:00 AM alert ("12:00 AM Community Board open!!") through the existing notification service, sent in batches
4. **Reads don't depend on the job.** `GET /boards/today` computes lock state from the server's clock and the user's timezone, so boards still show correctly if a job runs late; the job only triggers the alert.

Edge cases:

Case	Handling
Daylight saving time	Offsets are recalculated from `tz_iana` each day, so users move between offset groups automatically
Travel or timezone changes	The new timezone applies from the *next* local day. This stops users from changing timezones to get extra days or stamps.
Month end	The board closes at the end of `close_local` in each timezone; the **capsule is sealed** once the last timezone (UTC−12) has passed midnight
Retries	Unlock and notification jobs are **idempotent** (keyed by offset and date), so a retry never sends a second alert
Notification fatigue	One alert per day maximum; it respects the user's notification settings (PRD requirement 7.1)

3. Computing the Squad Stamp Tally

The problem: the tally ("Alex M. — 14 stamps") is read every time someone opens the Passport, but changes only when a stamp is earned. Counting every stamp on every read doesn't scale.

The approach: write-time counters, plus nightly reconciliation.

Step	How
1. Award	Tagging a post to a board writes `BOARD_POST` and `STAMP_AWARD` in **one database transaction**, then publishes `stamp.awarded`
2. Increment	The tally worker reads the event and increments a **Redis sorted set** per squad per month (`tally:{squad_id}:{yyyy-mm}`), with one score per member
3. Read	`GET /squads/{id}/tally` reads the sorted set: one fast lookup, already ranked
4. Reconcile	A **nightly job** recounts from Postgres, the source of truth, and repairs any drift from missed or duplicate events. This is the "refreshes nightly" requirement in PRD 7.5.

Tally rules:

- Only stamps earned **while in the squad** count; leaving the squad removes the member from that squad's tally.
- Ties rank by who reached the count first, so there are no shared ranks to explain.
- The tally **resets each month** with the boards, consistent with "no streak to lose."
- **No public or cross-squad leaderboard exists anywhere** (a PRD non-goal).

4. Enforcing squad-only privacy

The principle: privacy is enforced **on the server, on every read**. Hiding content in the app's interface isn't enough, because anyone calling the API directly would see everything.

Rule	Enforcement
Only squad members see a board post	Every feed, capsule and reaction query filters on the viewer being a current `SQUAD_MEMBER` of the post's `squad_id`. This happens in the data-access layer, so no endpoint can skip it.
Only opted-in members see a board	The viewer must also have an active `BOARD_MEMBERSHIP` for that board
Post to see (locked feed)	Squad posts for today stay blurred until the viewer has posted today. The server returns blurred placeholders, not the real images.
Photos can't be shared outside the app	Media is served through **short-lived signed URLs** issued only after the checks above; a copied link stops working within minutes
Leaving a squad, or opting out of a board	Access ends immediately; the member's tally score is removed
Deleting a post	The `BOARD_POST`, its reactions and its capsule entry are removed with it
Blocking	Blocked users never see each other's board posts, even in the same squad
Sponsors	Sponsors get **aggregate counts only** (posts and opt-ins per board), never photos, names or per-user data, and only when a board has 50+ participants, so no individual can be identified
No public metrics	Reactions are visible only inside the squad; no API returns public like counts

How we'd verify it: automated tests that call every board endpoint as a non-member and expect `403 Forbidden`, plus a privacy review before Phase 2 launch.

Rollout and reliability

Concern	Approach
Feature flag and kill switch	Boards sit behind a server-side flag; turning it off hides boards without touching regular posting
Experiment assignment	Assignment is keyed by **`squad_id`**, matching the squad-level randomization in the Test Plan
Analytics events	Emitted from the event bus (`board_optin`, `post_type_selected`, `stamp_post_success`, and others), as listed in the Test Plan
Failure isolation	If the Boards service fails, regular BeReal posting is unaffected; the app falls back to SEND
Performance targets	Tag-and-stamp under 300 ms at p95; Tally read under 50 ms at p95 *(planning targets)*

Open questions

- **Timezone source:** device setting only, or confirmed against location?
- **Squads and friend lists:** is a squad a subset of a user's friends, or a separate group with invitations?
- **Reaction storage:** reuse RealMoji infrastructure, or keep board reactions separate?
- **Capsule retention:** how long are sealed capsules kept, and can users export them?